HARSH RAJ

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Education

Northeastern University, Boston, MA — Dec 2026 (expected)
Master of Science in Computer Science — GPA: 3.8/4.0

Delhi Technological University, Delhi, India — August 2023
Bachelor of Technology in Engineering Physics (Minor in Computer Science) — GPA: 3.4/4.0

Technical Skills

Programming Languages: Python, SQL, JavaScript, C++, Java
Cloud and DevOps: AWS, GCP, Docker, Kubernetes, Git, CI/CD
Databases: PostgreSQL, MySQL, MongoDB, Redis, Vector Databases
Tools and Libraries: PyTorch, Jax, Hugging Face, Ray, LangChain, Pandas, NumPy, CUDA, Triton

Professional Experience

Machine Learning Research Intern, Scale AI, Manhattan, NYC, NY — May 2026 – Sept 2026
- Research on AI Agent Failure Taxonomy.
- **Impact**: Two works, covering novel Failure Taxonomy and Root-Cause Attribution of failures.

Research Engineer Intern, Bespoke Labs AI, Mountain View, CA — Jan 2026 – Apr 2026
- Curated synthetic RL training environments for leading frontier labs.
- **Impact**: Work contributed to some of the best proprietary CLI agents in the market.

Research Assistant, Forschungszentrum Jülich — July 2025 – Present
- Contribute to EU Open Foundational Model projects (MINERVA, OpenEuroLLM, ELLIOT) on JUPITER, EU's largest supercomputer.
- Build high-quality permissive datasets for LLMs, with data processing pipelines for mixing, formatting, upsampling, interleaving, and quality validation.
- **Impact**: Released MixtureVitae-300BT achieving 10% performance improvement over existing permissive datasets, also matching non-permissive alternatives.

Applied Scientist, Vijil AI — Aug 2023 – Nov 2024
- Contributed to vijil-evaluation platform for enterprise LLM workflow red-teaming. Integrated 15+ benchmark datasets for vulnerability assessment
- Developed vijil-fuzzer framework for adversarial text mutation, achieving 85% success rate against GPT-4o at launch
- **Impact**: The vijil-evaluate service is used by enterprise clients including DigitalOcean and Groq

Selected Publications

- Model or Harness? An Interaction-Centric Taxonomy for Localizing Agent Failures **1st Author** — *preprint*
- Consistency as a Testable Property: Statistical Methods to Evaluate AI Agent Reliability **1st Author** — NeurIPS 2026
- Terminal-Bench: Benchmarking Agents on Hard, Realistic Tasks in Command Line Interfaces **Core Team** — ICLR 2026
- MixtureVitae: Web-Scale Pretraining Dataset With High Quality Instruction and Reasoning **1st Author** — ICML 2026
- Measuring Reliability of Large Language Models through Semantic Consistency **1st Author** — NeurIPS 2022
  Best Paper Award

Projects

terminal-bench (2500+ GitHub ⋆) — Industry standard Benchmark for CLI Agents. **(Core Team)**
harbor (6000+ GitHub ⋆) — Language Agent Sandbox Evaluation framework. **(Core Team)**
inspect-ai (3000+ GitHub ⋆) — LLM Evaluation Suite. **(Contributor)**
garak (5000+ GitHub ⋆) — LLM Red-Teaming Suite. **(Contributor)**

Awards and Achievements

- SWISS AI Grant Recipient – Awarded 200k GPU hours and CHF 61,000 in funding — 2025
- 3rd Place – Vision and Language Navigation R2R Benchmark Competition — *2024*
- ML Safety Challenge – Best Paper Award with a $5000 prize — *2022*
- Rank 7 – Codeforces Raif ML Round 1 Competition — *2021*